

A THEORETICAL ANALYSIS

We analyze the main objectives in Secs. 4.1–4.3 from three aspects: the statistical meaning of the authenticity signal, the risk prioritization induced by the hierarchical adversarial reward, and the geometric effect of the cluster-driven objective. The analysis characterizes the objectives optimized by T-Rex rather than claiming global convergence of the practical RL procedure.

A.1 Authenticity-Aware Exploration

The authenticity discriminator $S_\psi : \mathcal{Z} \rightarrow [0, 1]$ is trained to distinguish latent embeddings of observed data from constructed negative samples. Let $p_+(z)$ and $p_-(z)$ denote the corresponding latent distributions with class priors π_+ and π_- , where $\pi_+, \pi_- > 0$. We consider latent points satisfying $\pi_+ p_+(z) + \pi_- p_-(z) > 0$.

LEMMA 1 (AUTHENTICITY LIKELIHOOD RATIO). *At the population optimum, $S_\psi^*(z) = \frac{\pi_+ p_+(z)}{\pi_+ p_+(z) + \pi_- p_-(z)}$. On the common support, $\text{logit } S_\psi^*(z) = \log \frac{p_+(z)}{p_-(z)} + \log \frac{\pi_+}{\pi_-}$; hence $R_{\text{auth}}(z) = \gamma_{\text{auth}} S_\psi^*(z)$ preserves the likelihood-ratio ordering for $\gamma_{\text{auth}} > 0$.*

PROOF. The population binary cross-entropy objective is

$$\mathcal{L}_{\text{auth}}(S) = -\pi_+ \int p_+(z) \log S(z) dz - \pi_- \int p_-(z) \log(1-S(z)) dz.$$

Since the objective is pointwise separable, for each fixed z it suffices to minimize

$$-\pi_+ p_+(z) \log S - \pi_- p_-(z) \log(1-S).$$

On the common support where $p_+(z) > 0$ and $p_-(z) > 0$, differentiating with respect to S gives

$$-\frac{\pi_+ p_+(z)}{S} + \frac{\pi_- p_-(z)}{1-S}.$$

Setting the derivative to zero yields

$$S_\psi^*(z) = \frac{\pi_+ p_+(z)}{\pi_+ p_+(z) + \pi_- p_-(z)}.$$

The pointwise objective is strictly convex for $S \in (0, 1)$ because

$$\frac{\pi_+ p_+(z)}{S^2} + \frac{\pi_- p_-(z)}{(1-S)^2} > 0,$$

so this stationary point is the unique minimizer. If $p_-(z) = 0$ and $p_+(z) > 0$, the minimum is attained at $S_\psi^*(z) = 1$; similarly, if $p_+(z) = 0$ and $p_-(z) > 0$, it is attained at $S_\psi^*(z) = 0$. These boundary cases agree with the same expression.

On the common support,

$$\frac{S_\psi^*(z)}{1-S_\psi^*(z)} = \frac{\pi_+}{\pi_-} \frac{p_+(z)}{p_-(z)},$$

and therefore

$$\text{logit } S_\psi^*(z) = \log \frac{p_+(z)}{p_-(z)} + \log \frac{\pi_+}{\pi_-}.$$

Since the sigmoid function is strictly increasing and $\gamma_{\text{auth}} > 0$, both $S_\psi^*(z)$ and $R_{\text{auth}}(z) = \gamma_{\text{auth}} S_\psi^*(z)$ preserve the ordering induced by $p_+(z)/p_-(z)$. $\square$

Lemma 1 gives R_{auth} a direct statistical interpretation. At the population optimum, it ranks latent candidates according to their relative support under the observed-data latent distribution versus the constructed negative distribution. It should therefore be interpreted as a relative plausibility signal rather than an exact test of membership in the unknown data manifold.

Writing $\sigma(\cdot)$ for the sigmoid function, this relation can also be expressed on the common support as

$$R_{\text{auth}}(z) = \gamma_{\text{auth}} \sigma\left(\log \frac{p_+(z)}{p_-(z)} + \log \frac{\pi_+}{\pi_-}\right),$$

which makes explicit how relative plausibility enters the exploration objective.

A.2 Confidence-Guided Risk Exploration

For a successful failure $\hat{y} \neq y^*$, Sec. 4.2 defines $R_{\text{adv}}(x, y^*) = \alpha_{\text{bonus}}(p)p$, where $p = P_f(\hat{y} | x)$ and $\alpha_{\text{bonus}}(p)$ follows Eq. (5). For $\hat{y} = y^*$, $R_{\text{adv}}(x, y^*) = -\beta_{\text{fail}} P_f(y^* | x)$, where $\beta_{\text{fail}} > 0$.

PROPOSITION 2 (MARGIN INTERPRETATION OF R_{adv}). *For $\hat{y} \neq y^*$, with $p = P_f(\hat{y} | x)$ and $M_f(x) = P_f(\hat{y} | x) - P_f(y^* | x)$, $M_f(x) \geq \max\{0, 2p - 1\}$, and the reward tiers in Eq. (5) are non-overlapping. For $\hat{y} = y^*$, R_{adv} strictly decreases with $P_f(y^* | x)$.*

PROOF. For a successful failure, write $p = P_f(\hat{y} | x)$ and $q = P_f(y^* | x)$. Since $\hat{y} \neq y^*$ and class probabilities sum to one, $p + q \leq 1$, and hence $q \leq 1 - p$. Therefore,

$$M_f(x) = p - q \geq 2p - 1.$$

Moreover, because $\hat{y}$ is the predicted class, $p \geq q$, which also gives $M_f(x) \geq 0$. Combining the two inequalities yields

$$M_f(x) \geq \max\{0, 2p - 1\}.$$

Consequently, $p > 0.5$ guarantees a positive probability margin over the intended label, while $p > 0.7$ guarantees $M_f(x) > 0.4$.

It remains to relate this margin guarantee to the hierarchical reward. Let $g(p) = \alpha_{\text{bonus}}(p)p$. From Eq. (5),

$$g(p) = \begin{cases} 0.5p, & p \leq 0.5, \\ p, & 0.5 < p \leq 0.7, \\ 2p, & p > 0.7. \end{cases}$$

Over the three confidence intervals, the corresponding rewards satisfy

$$g(p) \in \begin{cases} [0, 0.25], & p \leq 0.5, \\ (0.5, 0.7], & 0.5 < p \leq 0.7, \\ (1.4, 2], & p > 0.7. \end{cases}$$

Thus the three tiers are strictly ordered and non-overlapping. Within each tier, g also has a positive slope. Hence higher-confidence failures receive larger rewards. At the same time, the lower bound $\max\{0, 2p - 1\}$ is nondecreasing in p and becomes strictly increasing for $p > 0.5$; in particular, the high-confidence tier guarantees $M_f(x) > 0.4$.

For an unsuccessful candidate, letting $q = P_f(y^* | x)$ gives

$$R_{\text{adv}} = -\beta_{\text{fail}} q.$$

Since $\beta_{\text{fail}} > 0$, $q_1 > q_2$ implies

$$R_{\text{adv}}(q_1) < R_{\text{adv}}(q_2).$$

□

Proposition 2 strengthens the interpretation of the hierarchical reward beyond simple monotonicity. A larger successful-failure confidence receives a larger reward and yields a no weaker lower bound on the probability margin between the incorrect prediction and the intended label. In particular, the high-confidence tier guarantees a nontrivial target-label margin rather than treating all misclassifications equally.

Together with Lemma 1, the Phase-1 objective $R_{\text{base}} = R_{\text{auth}} + R_{\text{adv}}$ can therefore be interpreted as combining relative data plausibility with confidence-sensitive failure exploration equipped with a prediction-margin guarantee.

A.3 Cluster-Driven Region Formation

Phase 1 explores high-reward failures without explicitly constraining their spatial organization. Phase 2 introduces R_{cluster} in Eq. (6) to impose region-level structure.

For $K \geq 2$, let

$$C_k = \{z_i \in \mathcal{B}_t : c(i) = k\}, \quad n_k = |C_k| > 0,$$

and let

$$\mu_k = \frac{1}{n_k} \sum_{z_i \in C_k} z_i$$

be the corresponding K-Means centroid. Define

$$W(\mathcal{B}_t) = \frac{1}{B} \sum_{z_i \in \mathcal{B}_t} \|z_i - \mu_{c(i)}\|_2^2$$

and

$$D(\mathcal{B}_t) = \min_{j \neq k} \|\mu_j - \mu_k\|_2^2.$$

Then

$$R_{\text{cluster}}(\mathcal{B}_t) = -W(\mathcal{B}_t) + \lambda_{\text{sep}} D(\mathcal{B}_t), \quad \lambda_{\text{sep}} > 0.$$

When referring below to a global maximizer of R_{cluster} , we assume that the maximum is attained over the feasible set of batches.

PROPOSITION 3 (GEOMETRY OF R_{cluster}). *For each C_k, μ_k minimizes its squared within-cluster distortion, and $W(\mathcal{B}_t) = \frac{1}{B} \sum_{k=1}^K \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2$. If a feasible $\tilde{\mathcal{B}}_t$ satisfies $W(\tilde{\mathcal{B}}_t) \leq \eta$, $D(\tilde{\mathcal{B}}_t) \geq m^2$, and $\lambda_{\text{sep}} m^2 > \eta$, then any global maximizer $\mathcal{B}_t^*$ satisfies $D(\mathcal{B}_t^*) \geq m^2 - \eta/\lambda_{\text{sep}} > 0$.*

PROOF. For any u and cluster C_k ,

$$\begin{aligned} \sum_{z_i \in C_k} \|z_i - u\|_2^2 &= \sum_{z_i \in C_k} \|(z_i - \mu_k) + (\mu_k - u)\|_2^2 \\ &= \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2 + n_k \|u - \mu_k\|_2^2 \\ &\quad + 2(\mu_k - u)^\top \sum_{z_i \in C_k} (z_i - \mu_k). \end{aligned}$$

By the definition of μ_k ,

$$\sum_{z_i \in C_k} (z_i - \mu_k) = 0,$$

and therefore

$$\sum_{z_i \in C_k} \|z_i - u\|_2^2 = \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2 + n_k \|u - \mu_k\|_2^2.$$

The second term is nonnegative and vanishes only at $u = \mu_k$, so μ_k minimizes the squared within-cluster distortion.

For the pairwise distances within C_k , where the double sum is over all ordered pairs $(z_i, z_j) \in C_k \times C_k$,

$$\begin{aligned} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2 &= \sum_{i, j} \|(z_i - \mu_k) - (z_j - \mu_k)\|_2^2 \\ &= 2n_k \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2, \end{aligned}$$

where the cross term vanishes because the centered vectors sum to zero. Hence

$$\sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2 = \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2.$$

Summing over k and dividing by B gives

$$W(\mathcal{B}_t) = \frac{1}{B} \sum_{k=1}^K \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2.$$

Thus $W(\mathcal{B}_t)$ is exactly the normalized within-cluster pairwise squared dispersion, and maximizing the cohesion component $-W(\mathcal{B}_t)$ is therefore equivalent to minimizing this dispersion.

Now let $\mathcal{B}_t^*$ be a global maximizer of R_{cluster} . Since $\tilde{\mathcal{B}}_t$ is feasible,

$$R_{\text{cluster}}(\mathcal{B}_t^*) \geq R_{\text{cluster}}(\tilde{\mathcal{B}}_t) \geq -\eta + \lambda_{\text{sep}} m^2.$$

On the other hand, $W(\mathcal{B}_t^*) \geq 0$ gives

$$R_{\text{cluster}}(\mathcal{B}_t^*) = -W(\mathcal{B}_t^*) + \lambda_{\text{sep}} D(\mathcal{B}_t^*) \leq \lambda_{\text{sep}} D(\mathcal{B}_t^*).$$

Combining the two inequalities yields

$$D(\mathcal{B}_t^*) \geq m^2 - \frac{\eta}{\lambda_{\text{sep}}}.$$

Since $\lambda_{\text{sep}} m^2 > \eta$, the right-hand side is strictly positive. Hence the optimum retains a positive minimum inter-centroid separation and cannot exhibit complete centroid collapse. □

Proposition 3 gives complementary interpretations of the two terms in Eq. (6). Maximizing the cohesion term is exactly equivalent to reducing normalized within-cluster pairwise dispersion. For a global optimum of the clustering objective, the separation term further provides a quantitative lower bound on the minimum centroid distance whenever a sufficiently compact and separated feasible configuration exists.

Substituting the pairwise identity into the clustering contribution of Eq. (7) gives the exact reformulation

$$\lambda_{\text{cluster}} R_{\text{cluster}}(\mathcal{B}_t) = -\frac{\lambda_{\text{cluster}}}{B} \sum_{k=1}^K \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2 + \lambda_{\text{cluster}} \lambda_{\text{sep}} D(\mathcal{B}_t).$$

Thus Phase 2 augments the Phase-1 risk-authenticity objective with an explicit penalty on within-region pairwise dispersion and a reward on the minimum separation between generated regions. The proposition characterizes the geometry induced by the clustering component in latent space; it does not by itself imply topological connectedness, semantic equivalence in the original tabular feature space, or global convergence of the full Phase-2 RL objective.